

Borhan Javadian

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Research Interests

Computer vision and representation learning, mainly generalized category discovery and domain generalization, along with self-supervised learning and vision-language models. I also work on high-performance computing, in particular GPU programming and sparse linear algebra.

Education

Sabancı University 2023 – June 2027 (expected)
B.Sc. in Computer Science and Engineering Istanbul, Türkiye

- CGPA 3.82/4.00.
- Dean's Honor or High Honor List in every semester.
- Academic Success Scholarship (50% tuition) for 2025–26 and 2026–27, raised from a 25% admission scholarship.
- Courses: Algorithms, Data Structures, Parallel Computing, Digital Image and Video Analysis, Cryptography, Computer and Network Security, Computer Networks, Cloud Computing.

Sabancı University 2023 – June 2027 (expected)
Minor in Mathematics Istanbul, Türkiye

- GPA 4.00/4.00, with an A in every course taken.
- Courses: Linear Algebra, Probability, Mathematical Analysis, Abstract Algebra, Differential Equations, Discrete Mathematics, Elementary Number Theory.

Research Experience

Domain Generalization for Generalized Category Discovery in Remote Sensing Fall 2026 – present
Graduation project, Sabancı University Prof. Erchan Aptoula

- Working on generalized category discovery when test images come from domains the model never saw during training, with satellite imagery as the main application.
- Wrote a GCD pipeline in PyTorch with DINO features, contrastive training, k-means and semi-supervised k-means, and Old/New/All accuracy using Hungarian matching.
- Reproducing DG2CD-Net, HiDISC and other recent GCD and DG-GCD methods, and running them on the same datasets and evaluation settings so the results can be compared directly.
- In the second half of the project I will build a new method on top of these baselines.

Sparse Matrix-Vector Multiplication (SpMV) Summer 2026 – present
Research project, Sabancı University Prof. Kamer Kaya

- Implemented SpMV as sequential CSR, OpenMP, CUDA, MPI, and hybrid CPU-GPU versions.
- Compared matrix reorderings (RCM, AMD, GPS, and graph-based methods) and how much each one changes performance.
- Now benchmarking newer SpMV methods in one shared setup, across different sparse matrices, thread counts, and execution settings.

Fully Homomorphic Encryption for Matrix Multiplication July 2025 – Feb. 2026
PURE research intern, later research assistant, Sabancı University Prof. Erkay Savaş

- Implemented Halevi-Shoup and bicyclic encoding for matrix multiplication under CKKS, in C++ with Microsoft SEAL.
- Measured runtime, number of ciphertext rotations, and multiplicative depth for different matrix sizes, and tried batching and chunking for encrypted matrix-vector products.
- Presented the work at the Sabancı University PURE Research Symposium.

Selected Projects

Generalized Category Discovery

GCD on CIFAR-10 in PyTorch: DINO features, contrastive losses, k-means, semi-supervised k-means, and clustering evaluation.

SpMV Optimization

SpMV in C++ with sequential and OpenMP kernels, blocked methods, matrix reorderings, and benchmark plots.

Contree-Parallel

Building optimal classification trees in parallel with branch-and-bound, on both CPU and GPU.

ECC Signal Protocol

A Signal-style messaging protocol with elliptic-curve key exchange, Ed25519 signatures, and forward secrecy.

Student Stress Level Analysis

[code](#)

Statistical tests and classifiers (Random Forest, SVM, KNN, logistic regression, XGBoost) on how sleep, caffeine, fatigue and study habits relate to student stress.

Teaching

Learning Assistant, Sabancı University

2024 – present

Labs, recitations, office hours, grading, and exams

- CS 306 Database Systems (Fall 2026)
- CS 438 Blockchain: Security and Applications (Spring 2025)
- CS 303 Logic and Digital System Design (Fall 2024, Spring 2025)
- MATH 101 Calculus I (Spring 2024)

Honors and Awards

- Ranked first in Computer and Network Security.
- Highest score in the class in Parallel Computing, Blockchain: Security and Applications, and Elementary Number Theory.
- Advanced Group, inzva Algorithm Competition Summer Camp, 2026.
- 1st Place and Gold Medal, Caribou Mathematics Contest, 2021.
- Distinction, Fermat Mathematics Contest, University of Waterloo, 2021.

Industry Experience

Cybersecurity Intern, Proteo Cyber Defence AB

Summer 2026

Sweden

Remote

- Did attack-surface analysis and vulnerability assessments with DNS and subdomain enumeration, web crawling, TLS inspection, port scanning, CVE checks, and OSINT.
- Also worked on EDR/XDR, incident response, and detection engineering.

Skills

Programming	Python, C++ , C, CUDA, Go, SQL, JavaScript, Bash, $\LaTeX$
Machine learning	PyTorch, torchvision, scikit-learn, DINO/DINOv2, Vision Transformers, contrastive learning, clustering
Parallel computing	CUDA, OpenMP, MPI, SLURM, distributed-memory programming, sparse linear algebra
Cryptography	CKKS, Microsoft SEAL, elliptic-curve cryptography
Systems	Linux, Git, AWS, PostgreSQL, computer networks

Languages

Persian (native), English (C1), Turkish (Fluent)